

K-FIELD SOLAR-SYSTEM COUPLING ANALYSIS

Volume-Filling Cell3 Origin Lock, Daily Body-CMB Kinematics, Body-Sun Differential Force, and the Phase-Shifted Orbital Response

Principal Investigator: P. Dwayne Esterline

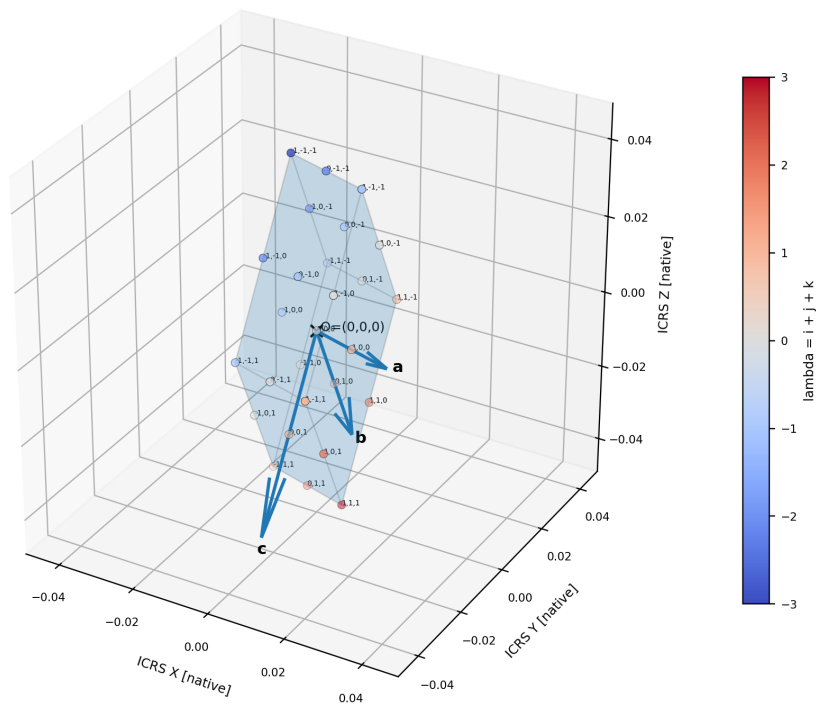
BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008)

Krytronx

Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

Generated: Unix epoch 1784823303 | UTC 2026-07-23T16:15:03Z

Origin-locked ICRS Cell3 base cell: 27 nodes and six boundary planes



Title figure. The Cell3 base cell is centered at the explicit ICRS Cartesian origin $O=(0,0,0)$. The primitive basis vectors, all 27 centered half-step nodes, and all six boundary planes are rendered directly from the locked machine-readable geometry. Axes are in native Cell3 length units and use equal three-dimensional scale.

Abstract

This reference develops a daily, one-year solar-system computation of the K-Field force direction and its phase-shifted orbital response. The Cell3 translational gauge is fixed at the ICRS Cartesian origin $O=(0,0,0)$, while its primitive basis retains the previously locked ICRS orientation. For every body and every daily epoch from 2026-07-23 through 2027-07-23, the calculation constructs the body velocity relative to the Solar-System barycenter, adds the locked Solar-System CMB translation vector, applies the phase metric G_{phase} , forms the spin-coupled whole-body K acceleration, removes the mass-weighted common Solar-System acceleration, and then forms the body-Sun differential vector that alone perturbs the heliocentric orbit. The force is resolved into radial, tangential, and normal orbital components. The tangential component changes orbital energy and the osculating semimajor axis. For a nearly circular prograde orbit under a fixed inertial force, integration produces an exact quarter-cycle relation: maximum positive tangential force is followed approximately 90 degrees later by maximum positive semimajor-axis displacement. The daily computation resolves fitted force-to-semimajor-axis phase lags of 90.1685 degrees for Mercury, 89.9820 degrees for Venus, and 89.5997 degrees for Earth. A complete fixed-force orbit test shows that the induced aphelion direction lies 89.9952, 90.0212, and 90.0083 degrees ahead of the projected K-force direction for Mercury, Venus, and Earth, respectively. The fixed inertial component does essentially zero net work over a closed orbit; therefore, the persistent signature is not an indefinitely enlarged semimajor axis. It is a coherent eccentricity-vector, apsidal, inclination, and annual-sideband modulation. Earth is a special geometric case in the selected year: its nominal aphelion direction is only 2.5062 degrees from the projected body-Sun K-force direction, making the raw annual radius signal strongly covariant with ordinary eccentricity. The complete vector template, rather than radius alone, is required for an observational solve.

Field	Locked or computed value
Coordinate orientation	ICRS/J2000 Cartesian axes
Cell3 origin	$O=(0,0,0)$ ICRS
Daily interval	2026-07-23 through 2027-07-23; 366 epochs
Bodies	Sun plus Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
Whole-body coefficient	$k_i=(1/3) \omega_i \text{ cross } [G_{\text{phase}}(V_0+v_i, \text{SSB})]$
Heliocentric observable	$q_i=k_i-k_{\text{Sun}}$
Resolved inner-planet force-to-size lags	Mercury 90.1685 deg; Venus 89.9820 deg; Earth 89.5997 deg
Closed-orbit induced-aphelion lags	Mercury 89.9952 deg; Venus 90.0212 deg; Earth 90.0083 deg
Earth aphelion alignment	2.5062 deg from projected K-force direction
Machine-readable companion	K-Field_solar_system_coupling_analysis_phase_shifted_1784823303.json

Contents

Abstract	2
1. Purpose, physical question, and result hierarchy	4
2. Coordinate locks and the Cell3 origin at $O=(0,0,0)$ ICRS	4
2.1 Metric and plane data	6
2.2 Consequence of the origin lock	6
3. Source data, ephemeris construction, and one-year sampling	6
4. Whole-body K-Field acceleration	7
4.1 Internal strain term and zero direct torque	8
5. Common-mode removal and the body-Sun observable	8
6. Orbital-frame decomposition	8
7. Analytical origin of the 90-degree phase shift	9
7.1 Closed-orbit work and the surviving eccentricity vector	9

8. Daily numerical results	10
9. Inner-planet phase results	11
9.1 Mercury	11
9.2 Venus	11
9.3 Earth	12
9.4 Mars	12
10. Earth as the resolved annual case	12
11. Outer planets and one-year resolution limits	14
12. CMB vector and K-direction behavior	15
13. Predicted orbital observables	15
13.1 Semimajor-axis modulation	15
13.2 Eccentricity and apsidal modulation	16
13.3 Inclination and node	16
13.4 Frequency structure	16
14. Comparison with known orbit kinematics	16
15. Variational equation for a definitive ephemeris fit	16
16. Reproduction procedure	16
17. Numerical checks and interpretation guards	17
18. Principal findings	17
19. Conclusion	17
Appendix A. Complete 27-node Cell3 registry	18
Appendix B. Vertices and boundary-plane equations	18
Appendix C. Complete body summary	19
Appendix D. Selected daily vector ledger	19
Appendix E. Formula ledger	20
Appendix F. Source references	20
Appendix G. Machine-readable reconstruction metadata	20

1. Purpose, physical question, and result hierarchy

The physical question is whether a correctly oriented, volume-filling K-Field produces a body-indexed modulation of known Solar-System orbits. The decisive causal sequence is not total force magnitude alone. It is the signed tangential component of the body-Sun differential K acceleration, because only that component performs instantaneous work on the orbit. The sequence examined is:

body CMB velocity -> transformed K vector -> spin-coupled acceleration -> body-Sun differential -> tangential work -> velocity change -> orbit-size response

The response is separated into four levels. First, the absolute K acceleration of each rotating body is computed. Second, the common Solar-System acceleration is removed. Third, the body-Sun differential vector is projected into the local radial-transverse-normal orbital frame. Fourth, the resulting perturbation is integrated and compared with the known Keplerian radius and apsidal geometry. This ordering prevents common-mode acceleration from being mistaken for an internal orbital effect.

The central result is that the proposed quarter-cycle response is present in the governing equations and in the daily numerical calculation for bodies completing at least one orbit in the interval. The same calculation also shows why a fixed K vector does not continuously pump the semimajor axis: its positive and negative work cancel over a closed orbit. The retained geometric memory is carried primarily by the eccentricity vector and by smaller out-of-plane terms.

Resolved result: maximum positive tangential K acceleration is followed approximately 90 orbital degrees later by maximum positive osculating semimajor-axis displacement; the induced aphelion direction is also approximately 90 degrees from the projected fixed-force direction.

2. Coordinate locks and the Cell3 origin at O=(0,0,0) ICRS

The ICRS axes define orientation. The explicit origin lock introduced here defines the Cell3 translational gauge: the center of the authoritative base cell is placed at the Cartesian coordinate O=(0,0,0). The primitive basis vectors are not rotated or rescaled. They retain their source-locked ICRS components.

Basis vector	ICRS components [native]	Magnitude [native]
a	[0.0203814385169, -0.00341641449641, -0.00233296591673]	0.020797058782
b	[0.00219874596832, 0.0129021702587, -0.0340665015361]	0.036494205131
c	[-0.000630379485347, -0.0242577220171, -0.0429243545711]	0.049308565900

$$\mathbf{x}(i, j, k) = \mathbf{O} + i \mathbf{a} + j \mathbf{b} + k \mathbf{c}, \quad \mathbf{O} = [0, 0, 0]_{\text{ICRS}}$$

The centered 27-node registry uses i, j, k in $\{-1/2, 0, +1/2\}$. The boundary vertices use all eight sign combinations of the half-step coefficients. The infinite volume-filling lattice is generated by integer translations $m \mathbf{a} + n \mathbf{b} + p \mathbf{c}$. The origin choice therefore fixes which primitive cell contains the ICRS origin without changing the orientation, metric, face areas, or force tensor.

Origin-locked ICRS Cell3 base cell: 27 nodes and six boundary planes

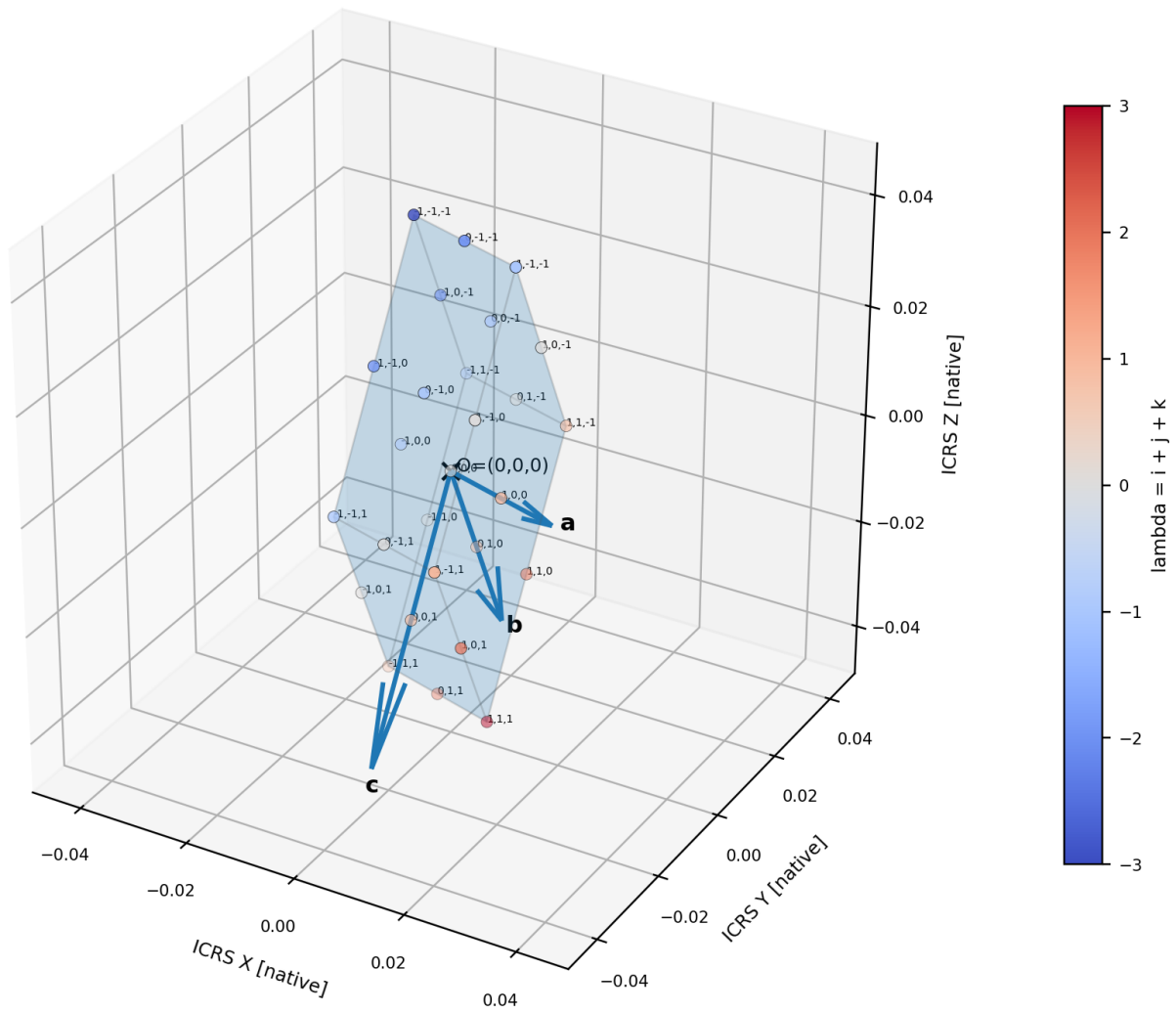


Figure 2.1. Dimensionally scaled ICRS Cell3 base geometry with $O=(0,0,0)$, all 27 nodes, primitive vectors, and the six boundary faces. Node labels are the integer doubled half-step indices $(2i, 2j, 2k)$.

Cell3 boundary-face node panels; origin O=(0,0,0) ICRS

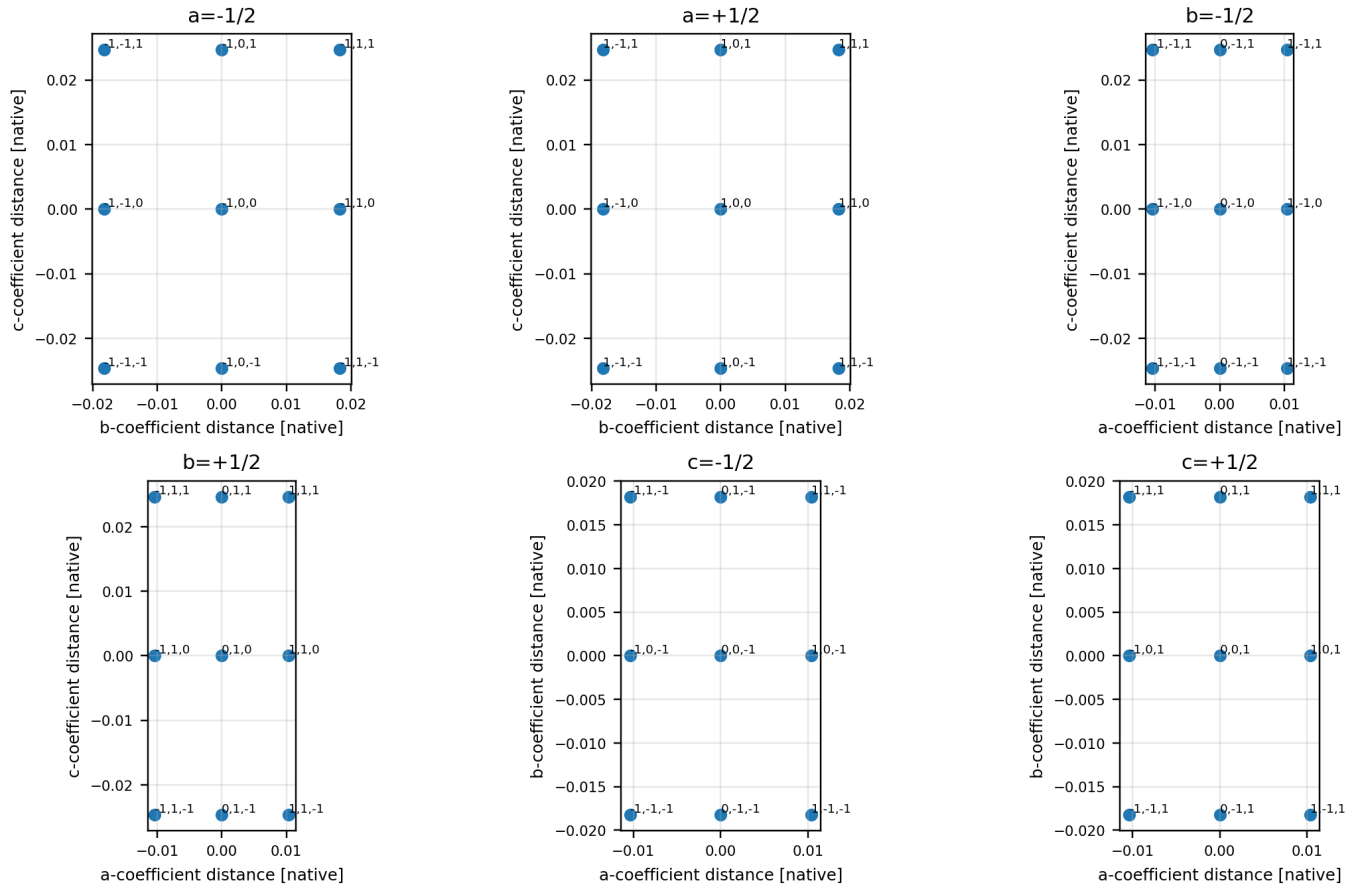


Figure 2.2. All six Cell3 boundary faces. Each panel contains the nine nodes lying on that face. In-plane axes use basis-coefficient distance in native units; the panels are therefore dimensionally tied to the primitive edge magnitudes.

2.1 Metric and plane data

Face family	Area [native^2]	Included angle [deg]	Unit normal ICRS
A_ab	7.547217896726240e-04	83.933483012	[0.1940922008, 0.9131771688, 0.3583792348]
A_ac	1.011255757688065e-03	80.448130018	[0.0890525941, 0.8665767693, -0.4910339483]
A_bc	1.385784431733287e-03	50.363231814	[-0.9959651903, 0.08360216251, -0.03261929056]

The primitive volume is 2.842066937809649e-05 native^3. The six boundary planes occur as three antipodal pairs. Their exact equations are tabulated in Appendix B.

2.2 Consequence of the origin lock

The present coupling equation uses the spatially uniform tensor G_{phase} . Under that equation, translating the Cell3 origin does not change the daily force directions because no position-dependent amplitude function has been inserted. The origin lock nevertheless completes the spatial indexing rule. Any later phase law $\Phi(\zeta)$, with $\zeta = B^{-1}x$ modulo the primitive lattice, can now be evaluated with x measured directly from $O=(0,0,0)$. This report does not invent such a phase-amplitude law; it evaluates the fully specified translation-invariant force equation and records the origin needed for the next spatially phased calculation.

3. Source data, ephemeris construction, and one-year sampling

The daily state calculation covers 366 inclusive UTC dates. Planetary heliocentric positions are generated from the JPL approximate Keplerian elements valid for 1800-2050, converted from the J2000 ecliptic to the ICRF-equatorial frame. Velocities are obtained by centered finite differentiation. The Sun-to-SSB offset is reconstructed by mass weighting the Sun and eight planetary-system positions using DE440 gravitational parameters. This method is adequate for degree-scale phase and direction screening. JPL identifies these formulae as lower-accuracy approximations and directs precision residual work to Horizons or integrated DE ephemerides [R1-R3].

Planetary spin poles and signed prime-meridian rates are taken from the NAIF generic planetary constants model. The signed rate is essential: retrograde Venus and Uranus reverse the spin vector and therefore reverse the cross-product branch relative to an unsigned-axis treatment [R4].

Source object	Use in computation	Resolution/status
JPL approximate elements 1800-2050	Daily heliocentric positions and orbital phase	Screening-grade analytic ephemeris
DE440 mass parameters	Mass-weighted SSB reconstruction	Published JPL system GMs
NAIF PCK pole and rotation model	Body spin vectors in ICRF	Generic ICRF orientation model
Locked G_phase and V0	K-Field transformed CMB vector	Source-locked numerical inputs
Cell3 geometry	ICRS orientation, basis, nodes, planes	Origin translated to O=(0,0,0) only

Daily heliocentric trajectories and normalized body-Sun K-direction arrows

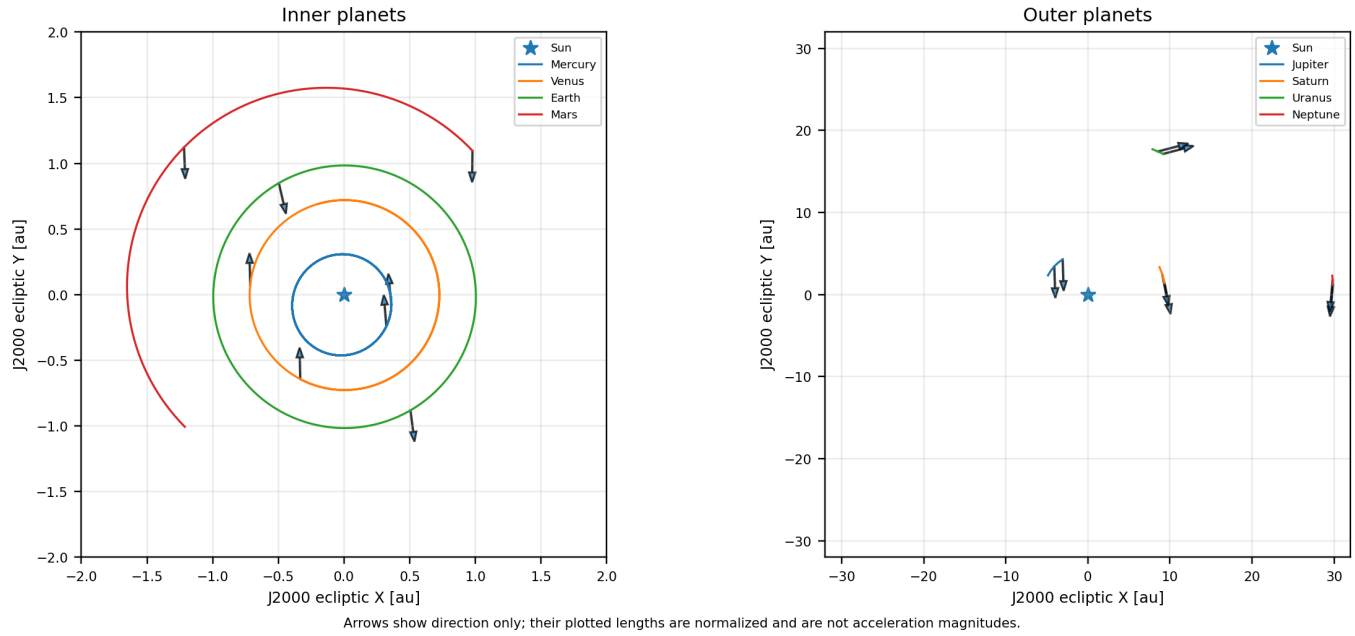


Figure 3.1. Daily heliocentric trajectories for the one-year interval. Arrows show the normalized body-Sun differential K direction at two epochs for each body. Orbit coordinates are actual computed J2000-ecliptic positions in astronomical units; arrow length is intentionally normalized and carries no magnitude scale.

4. Whole-body K-Field acceleration

For an extended rotating body, the local radial force density contains a translation-rotation cross term and a purely rotational quadratic term. Antipodal integration cancels the purely rotational term from the center-of-mass resultant. It remains as an internal radial stress modulation. The translation-coupled term survives.

$$a_{i,K} = \kappa_F H_i [\omega_i \text{ cross } G_{\text{phase}} V_{i,CMB}]$$

Here H_i is the dimensionless radial shape tensor. For a spherically symmetric mass distribution, $H_i = (1/3)I$, independent of the radial density profile. The body coefficient used in the daily calculation is therefore:

$$k_i(t) = (1/3) \omega_i(t) \text{ cross } \{ G_{\text{phase}} [V_0 + v_{i,SSB}(t)] \}$$

$$a_{i,K}(t) = \kappa_F k_i(t)$$

The body mass cancels from the acceleration. Mass remains relevant to the absolute force $F_{i,K} = M_i a_{i,K}$, but not to the orbit response per unit coupling. The force is orthogonal to both the spin vector and the transformed CMB vector at every epoch.

Quantity	Numerical value
G_phase row 1	[2.9764106378, 1.2431507795, -2.0803600354]
G_phase row 2	[1.2431507795, 2.5922286635, -2.3178839937]
G_phase row 3	[-2.0803600354, -2.3178839937, 6.7735709133]
V0, SSB relative to CMB [m/s]	[-359003.199066, 76679.886034, -44701.0601007]
G_phase V0 [m/s]	[-880221.981635, -143911.436562, 266335.026949]

Quantity	Numerical value
V0 [m/s]	369812.502000
G_phase V0 [m/s]	930825.217270

4.1 Internal strain term and zero direct torque

The canceled rotational self-term is not physically absent. It produces angle-dependent radial loading within each body. Because every elemental force is parallel or antiparallel to its local radius, $\mathbf{r} \times d\mathbf{F}$ is zero point by point. The equation therefore produces no direct spin torque, while still producing internal strain modulation and a translation-coupled center-of-mass acceleration.

5. Common-mode removal and the body-Sun observable

The absolute acceleration of the complete Solar System is not observable through internal ranging. Let M_i be the body or planetary-system masses used in the barycentric reconstruction. The common coefficient is:

$$\mathbf{k}_{SSB}(t) = [\sum_i M_i \mathbf{k}_i(t)] / [\sum_i M_i]$$

The barycentric acceleration of body i is $\mathbf{kappa}_F[\mathbf{k}_i - \mathbf{k}_{SSB}]$. For a heliocentric orbital measurement, the common term cancels again, leaving the exact body-Sun differential:

$$\mathbf{q}_i(t) = \mathbf{k}_i(t) - \mathbf{k}_{Sun}(t)$$

$$\mathbf{delta} \mathbf{a}_{iSun,K}(t) = \mathbf{kappa}_F \mathbf{q}_i(t)$$

This is the vector that must be compared with known orbit kinematics. Using the absolute body vector would incorrectly include acceleration shared with the Sun. Using only the Solar-System common mode would remove too much, because the Sun and each planet possess different spin axes and rotation rates.

Common-mode statistic	Value
Magnitude min [m/s ² per kappa]	0.932933377
Magnitude max [m/s ² per kappa]	0.933574364
Initial vector [m/s ² per kappa]	[0.02001600607, -0.8466008811, -0.3929725192]
Initial direction [RA, Dec]	271.354380 deg, -24.893523 deg

6. Orbital-frame decomposition

At every daily epoch, the body-Sun vector $\mathbf{q}_i$ is resolved in the local radial-transverse-normal frame. The radial unit vector $\mathbf{R}$ points from the Sun to the body. The normal unit vector $\mathbf{N}$ follows the orbital angular momentum. The transverse unit vector $\mathbf{T} = \mathbf{N} \times \mathbf{R}$ points in the prograde direction.

$$\mathbf{q}_R = \mathbf{q} \cdot \mathbf{R}, \quad \mathbf{q}_T = \mathbf{q} \cdot \mathbf{T}, \quad \mathbf{q}_N = \mathbf{q} \cdot \mathbf{N}$$

The three components have different dynamical roles. $\mathbf{q}_R$ directly changes the radial equation and contributes to eccentricity. $\mathbf{q}_T$ performs work because the unperturbed velocity is principally transverse; it changes orbital energy and the osculating semimajor axis. $\mathbf{q}_N$ changes inclination and node. The total force magnitude alone cannot identify where the orbit gains or loses energy.

$$d \epsilonpsilon / dt = \mathbf{v} \cdot (\mathbf{kappa}_F \mathbf{q}) \text{ approximately } \mathbf{v} \cdot \mathbf{kappa}_F \mathbf{q}_T$$

$$da/dt = [2 a^2 / \mu_{Sun}] \mathbf{v} \cdot (\mathbf{kappa}_F \mathbf{q})$$

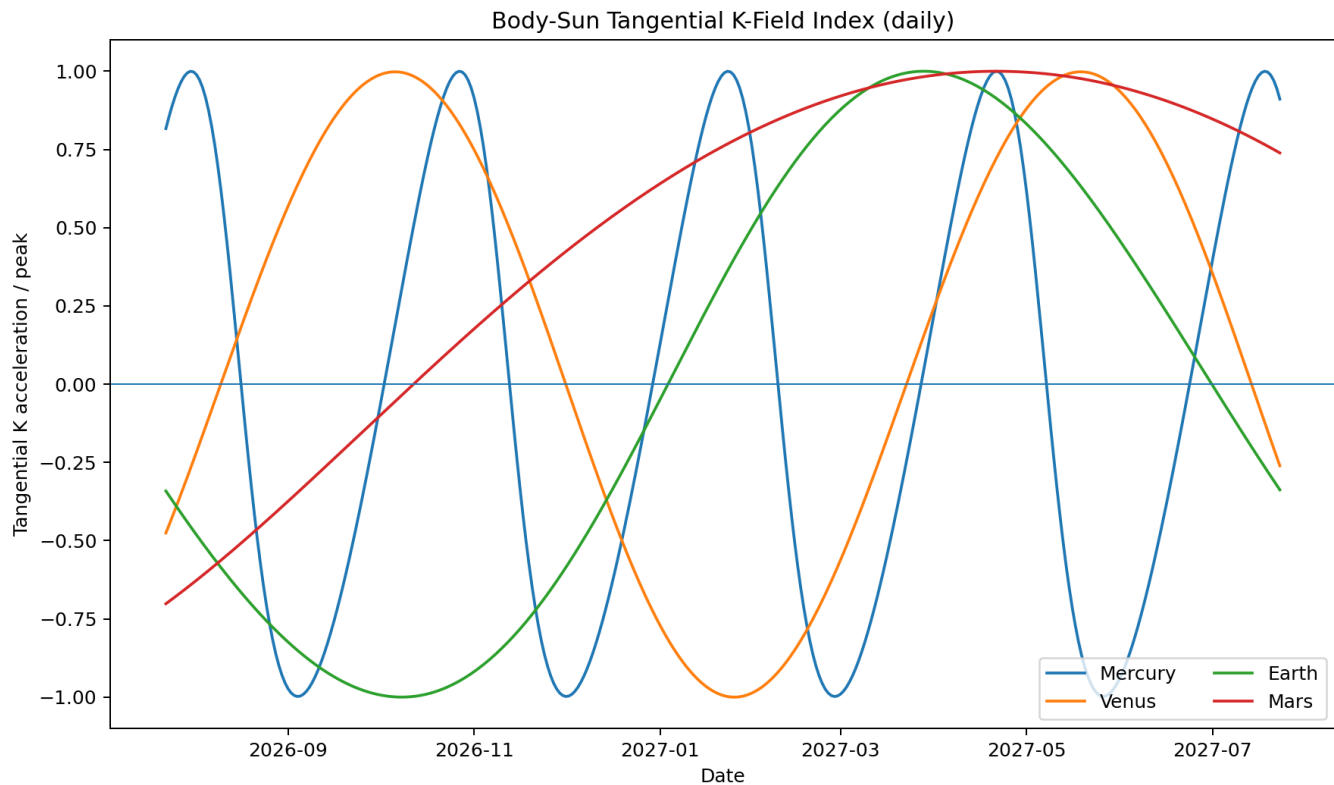


Figure 6.1. Daily normalized tangential K index for Mercury, Venus, Earth, and Mars. The different oscillation periods are the orbital periods; the sign identifies energy gain or loss.

7. Analytical origin of the 90-degree phase shift

Consider a nearly circular prograde orbit and a fixed K-force projection q_{parallel} in the orbital plane. Let $\theta=0$ at the radial direction aligned with q_{parallel} . The radial and transverse components are then:

$$q_R = q_{\text{parallel}} \cos(\theta), \quad q_T = -q_{\text{parallel}} \sin(\theta)$$

Maximum positive tangential acceleration occurs at $\theta=-90$ degrees. The semimajor-axis rate is proportional to q_T . Integrating with $d\theta/dt=n$ produces a cosine response:

$$\Delta a(\theta) = [2 \kappa_F q_{\text{parallel}} / n^2] \cos(\theta) + \text{integration constants}$$

The maximum positive semimajor-axis displacement occurs at $\theta=0$, one quarter orbit after the maximum positive tangential force. This is the precise mathematical form of the proposed force-to-velocity-to-enlarged-path sequence.

7.1 Closed-orbit work and the surviving eccentricity vector

A fixed inertial force is conservative over a closed geometric path in the limited sense that its line integral around the closed orbit vanishes: $\int q_{\text{dot}} dr = 0$. The semimajor-axis increase accumulated over one half orbit is returned over the other half. The persistent first-order result is a forced eccentricity-vector change. For a near-circular orbit:

$$\langle de/dt \rangle = -[3 \kappa_F / (2 n a)] \hat{h} \text{ cross } q_{\text{parallel}}$$

The eccentricity vector points toward perihelion. The induced aphelion points opposite it and therefore lies approximately $+90$ degrees from the projected K-force direction for a prograde orbit. This second 90-degree relation is distinct from, but geometrically consistent with, the tangential-force-to-semimajor-axis lag.

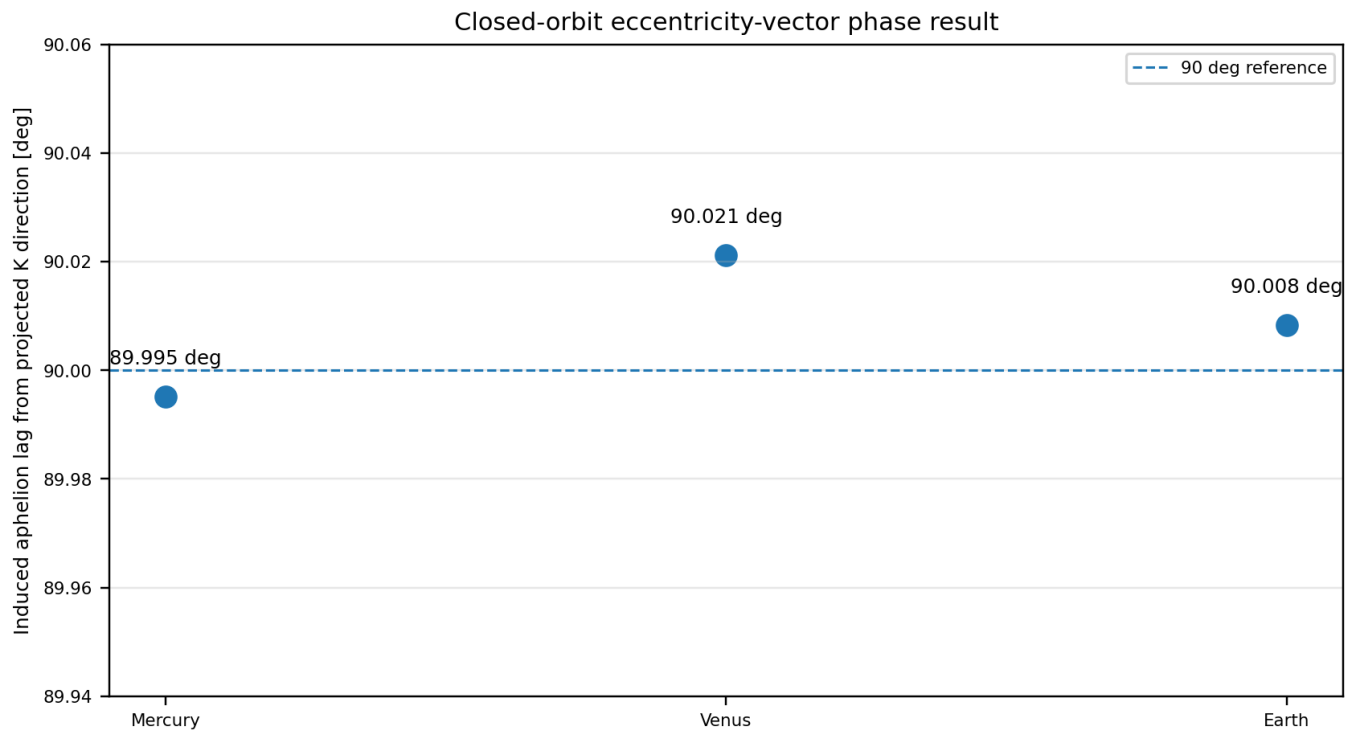


Figure 7.1. Complete fixed-force orbit tests for Mercury, Venus, and Earth. The induced aphelion direction closes at 90 degrees from the projected K-force direction to within about 0.02 degrees in the screening integration.

8. Daily numerical results

Body	Orbit covered [deg]	q min-max [$\text{m/s}^2/\kappa$]	Force $\rightarrow$ delta-a lag [deg]	Closed-orbit aphelion lag [deg]	Nominal aphelion to K [deg]
Mercury	1498.661	0.414 - 0.553	90.169	89.995	165.970
Venus	583.722	0.947 - 0.969	89.982	90.021	139.441
Earth	359.755	18.659 - 23.283	89.600	90.008	2.506
Mars	171.437	19.597 - 23.107	106.155	88.723	114.574
Jupiter	28.777	51.731 - 52.512	103.752	72.198	76.354
Saturn	12.552	49.197 - 49.600	72.359	93.622	7.882
Uranus	4.178	31.033 - 31.057	84.895	73.132	26.778
Neptune	2.212	32.985 - 33.014	88.321	93.999	45.651

Mercury, Venus, and Earth complete at least one full orbit in the one-year window, so their harmonic phase estimates are directly interpretable. Mars covers 171.4 degrees; the outer planets cover much smaller arcs. Their one-year fitted phase numbers are therefore partial-arc correlations rather than closed-orbit measurements. The complete fixed-force tests for those bodies must be run over their full orbital periods before applying the same closure standard.

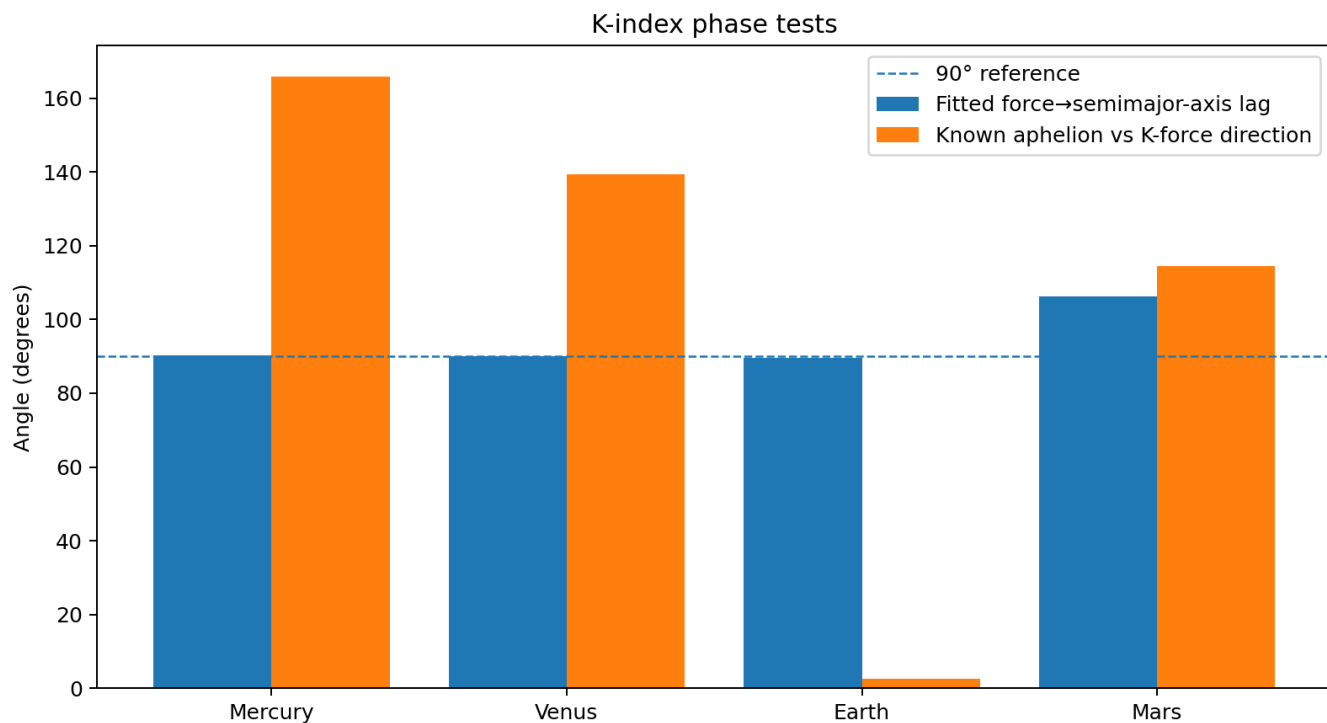


Figure 8.1. Phase tests from the daily calculation. The blue bars show the fitted tangential-force to semimajor-axis lag. The orange bars show the existing nominal aphelion direction relative to the K-force direction; this second quantity is not required to equal 90 degrees because it includes the pre-existing planetary eccentricity vector.

9. Inner-planet phase results

9.1 Mercury

Mercury covers 1498.660938 degrees during the interval. The body-Sun differential magnitude spans 0.413554651 to 0.552785331 m/s² per unit κ_F , and the tangential component spans -0.482100845 to 0.483136677 m/s² per unit κ_F . Maximum total magnitude occurs on 2026-10-03; maximum positive tangential force occurs on 2027-07-18. The fitted tangential-force to semimajor-axis phase lag is 90.168502 degrees. The complete fixed-force test gives an induced-aphelion lag of 89.995188 degrees from the projected K-force direction. The nominal pre-existing aphelion differs from the K direction by 165.970022 degrees.

Quantity	Value at $ \kappa_F =5.50e-18$
Periodic semimajor-axis half-amplitude [m]	7.9550715e-06
One-year $a \Delta e $ radial scale [m]	1.5315188e-04
Maximum causal radial perturbation [m]	1.4151685e-04
End-of-window causal radial perturbation [m]	-1.3279645e-04
K-direction excursion [deg]	3.885017
CMB-direction excursion [deg]	7.548634

9.2 Venus

Venus covers 583.722394 degrees during the interval. The body-Sun differential magnitude spans 0.947079142 to 0.969319799 m/s² per unit κ_F , and the tangential component spans -0.958945109 to 0.957245893 m/s² per unit κ_F . Maximum total magnitude occurs on 2027-07-16; maximum positive tangential force occurs on 2026-10-06. The fitted tangential-force to semimajor-axis phase lag is 89.982008 degrees. The complete fixed-force test gives an induced-aphelion lag of 90.021169 degrees from the projected K-force direction. The nominal pre-existing aphelion differs from the K direction by 139.440628 degrees.

Quantity	Value at $ \kappa_F =5.50e-18$
Periodic semimajor-axis half-amplitude [m]	1.0061308e-04
One-year $a \Delta e $ radial scale [m]	7.5306438e-04
Maximum causal radial perturbation [m]	5.2843920e-04
End-of-window causal radial perturbation [m]	3.8023824e-04

Quantity	Value at $ \kappa_F =5.50e-18$
K-direction excursion [deg]	0.390570
CMB-direction excursion [deg]	6.437150

9.3 Earth

Earth covers 359.755355 degrees during the interval. The body-Sun differential magnitude spans 18.659158033 to 23.283165589 m/s² per unit κ_F , and the tangential component spans -19.175911263 to 19.178801442 m/s² per unit κ_F . Maximum total magnitude occurs on 2027-01-02; maximum positive tangential force occurs on 2027-03-28. The fitted tangential-force to semimajor-axis phase lag is 89.599664 degrees. The complete fixed-force test gives an induced-aphelion lag of 90.008349 degrees from the projected K-force direction. The nominal pre-existing aphelion differs from the K direction by 2.506203 degrees.

Quantity	Value at $ \kappa_F =5.50e-18$
Periodic semimajor-axis half-amplitude [m]	0.005300372
One-year $a \Delta e $ radial scale [m]	0.0249705
Maximum causal radial perturbation [m]	0.02261952
End-of-window causal radial perturbation [m]	0.007315672
K-direction excursion [deg]	3.184745
CMB-direction excursion [deg]	4.648050

9.4 Mars

Mars covers 171.436956 degrees during the interval. The body-Sun differential magnitude spans 19.597311180 to 23.107222167 m/s² per unit κ_F , and the tangential component spans -14.756824082 to 21.035625193 m/s² per unit κ_F . Maximum total magnitude occurs on 2026-10-18; maximum positive tangential force occurs on 2027-04-21. The fitted tangential-force to semimajor-axis phase lag is 106.154934 degrees. The complete fixed-force test gives an induced-aphelion lag of 88.723368 degrees from the projected K-force direction. The nominal pre-existing aphelion differs from the K direction by 114.573693 degrees.

Quantity	Value at $ \kappa_F =5.50e-18$
Periodic semimajor-axis half-amplitude [m]	0.01735534
One-year $a \Delta e $ radial scale [m]	0.05347855
Maximum causal radial perturbation [m]	0.02509925
End-of-window causal radial perturbation [m]	-0.01360093
K-direction excursion [deg]	2.646234
CMB-direction excursion [deg]	4.066279

10. Earth as the resolved annual case

Earth provides the cleanest one-year phase test because the interval covers 359.755 degrees, nearly one complete sidereal orbit. The maximum total body-Sun K magnitude occurs near 2027-01-02. This is not the orbital-energy maximum. The maximum positive tangential K component occurs on 2027-03-28. The K-force projection becomes radially aligned on 2027-07-03, and the screening ephemeris places nominal aphelion on 2027-07-06.

Event	Date	Role
Maximum total body-Sun K magnitude	2027-01-02	Largest $ q $; not maximum work
Maximum positive tangential K force	2027-03-28	Maximum positive orbital-energy rate
K radial alignment	2027-07-03	Approximately 90 deg after tangential maximum
Nominal aphelion	2027-07-06	Known Keplerian maximum radius

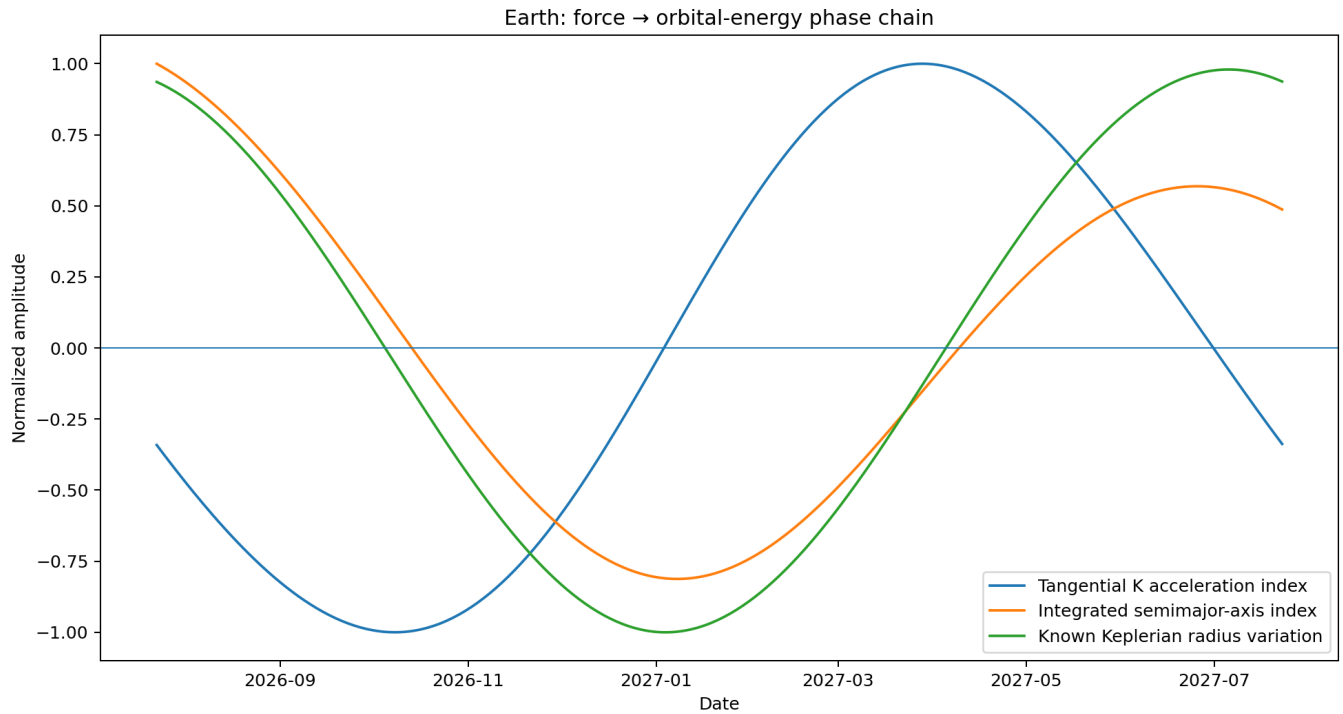


Figure 10.1. Earth force-to-energy phase chain. The tangential K acceleration is integrated into an osculating semimajor-axis response. The known Keplerian radius variation is shown as a separate normalized baseline. The curves demonstrate the quarter-cycle force-to-size relation and the near phase coincidence of the K radial direction with ordinary aphelion.

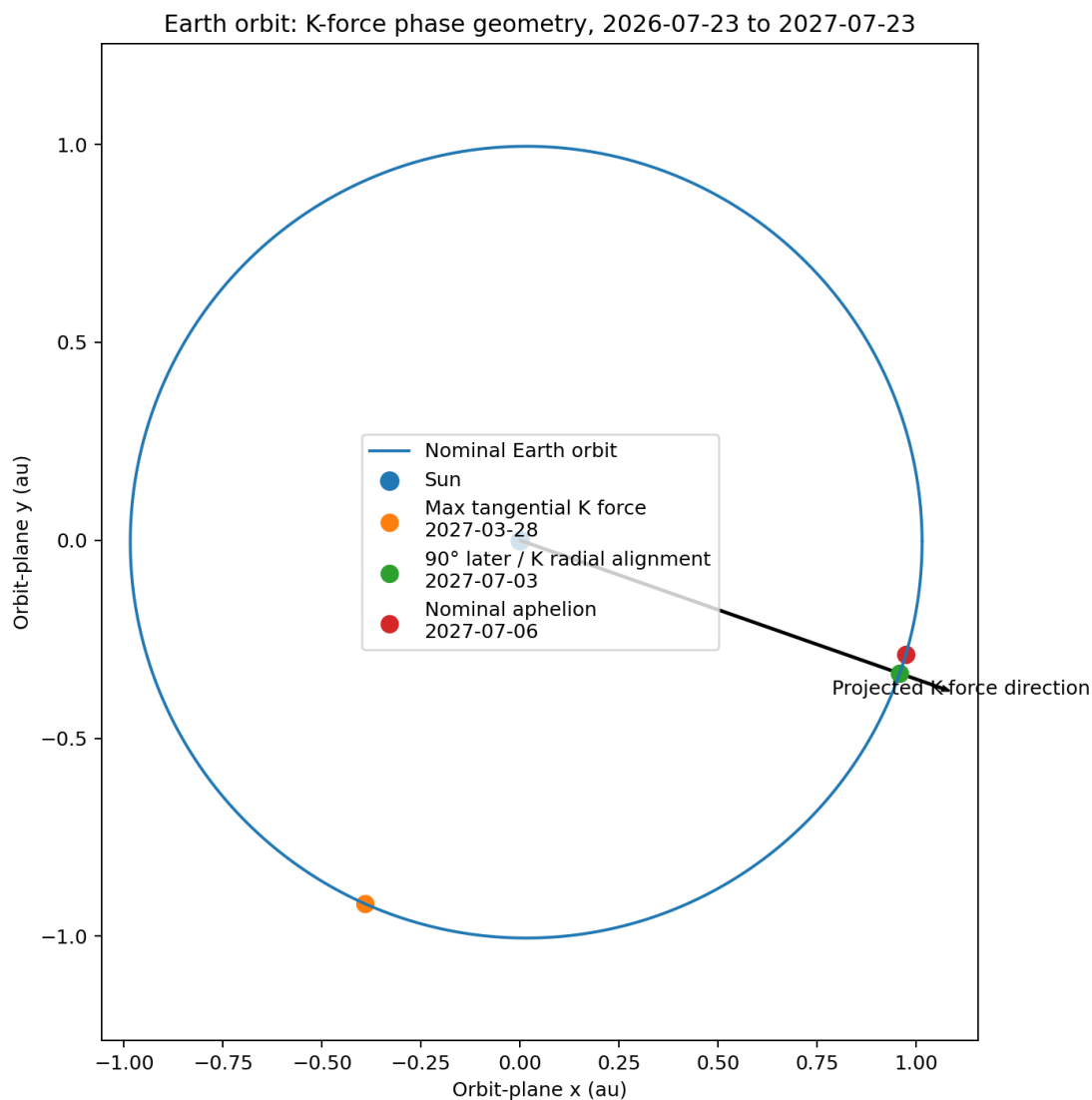


Figure 10.2. Earth orbital phase geometry. Maximum positive tangential K force precedes the radial K alignment by approximately one quarter orbit. The nominal aphelion follows the K radial-alignment point by only about three days in this interval.

Earth's nominal aphelion direction is separated from the projected body-Sun K direction by only 2.506203 degrees. This near alignment makes a simple annual radius comparison non-identifying: a K-indexed term can be absorbed strongly into fitted eccentricity magnitude, eccentricity-vector phase, and initial state. The discriminating observable is the complete vector response, including out-of-plane acceleration, annual CMB-vector wobble, and the different spin-indexed templates of other bodies.

11. Outer planets and one-year resolution limits

Jupiter, Saturn, Uranus, and Neptune traverse only 28.78, 12.55, 4.18, and 2.21 orbital degrees in one year. Their body-Sun K vectors are nearly stationary in inertial direction during the same interval. A one-year regression can estimate local slope and curvature but cannot determine a closed-orbit 90-degree response. The daily vectors remain useful because they specify the forcing template for a later multi-decade ephemeris fit.

Body	Coverage [deg]	q min-max	K direction excursion [deg]	CMB direction excursion [deg]	Nominal aphelion to K [deg]
Jupiter	28.77707	51.7309 - 52.5123	0.211813	0.316778	76.3537
Saturn	12.55167	49.1969 - 49.5999	0.089082	0.080081	7.8821
Uranus	4.17774	31.0329 - 31.0574	0.007461	0.036081	26.7779
Neptune	2.21155	32.9849 - 33.0142	0.002138	0.004871	45.6506

Saturn has a nominal aphelion-to-K separation of 7.882 degrees, the second-smallest value after Earth. The other bodies do not share this

alignment. Therefore, the Earth and Saturn alignments are geometric features to test, not a universal rule imposed by the current force equation.

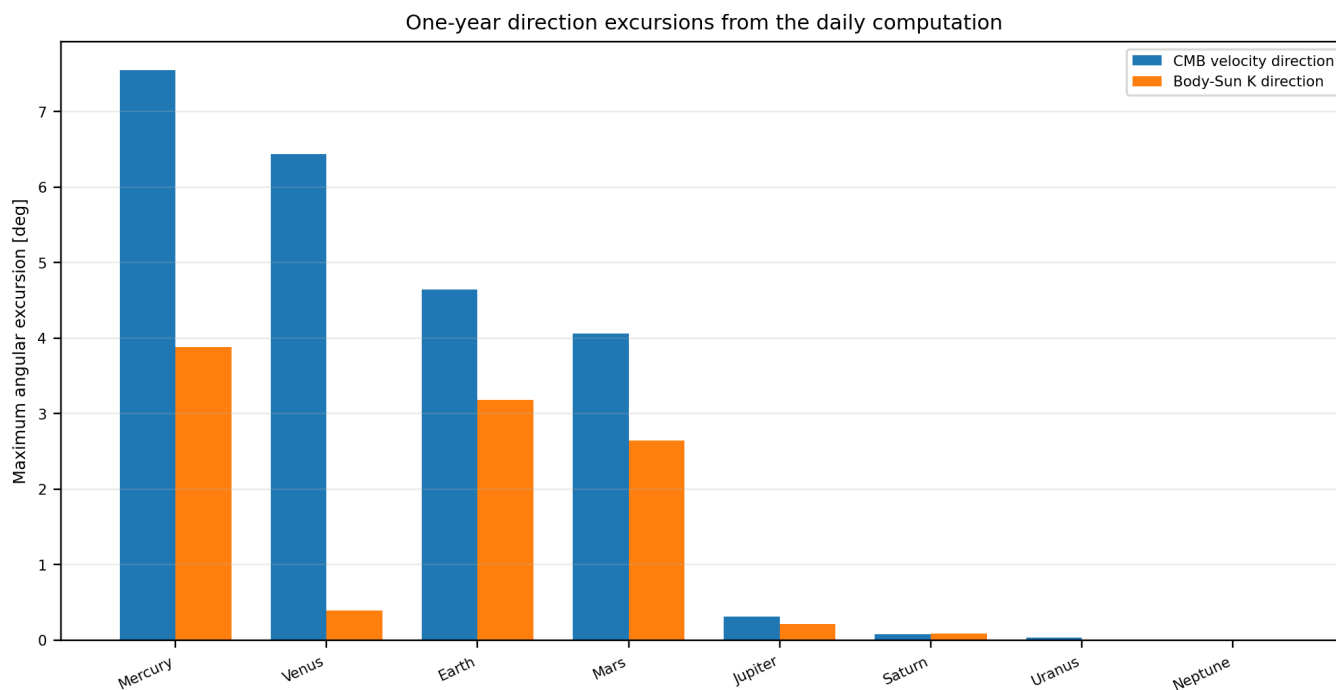


Figure 11.1. One-year angular excursions of the body CMB velocity direction and the body-Sun differential K direction. The excursions decrease strongly with heliocentric orbital speed and orbital period.

12. CMB vector and K-direction behavior

Each body possesses a distinct CMB velocity because its SSB velocity is added to the common translation vector V_0 . The transformed vector $G_{\text{phase}} V_{i,\text{CMB}}$ therefore varies in both magnitude and direction. The spin cross product converts this transformed CMB state into a body-indexed force direction. Fast inner-planet orbital velocities produce the largest annual direction excursions; slow outer-planet velocities produce nearly fixed directions.

Body	CMB speed min-max [m/s]	CMB direction excursion [deg]	Absolute K direction excursion [deg]	Body-Sun K excursion [deg]
Mercury	332794.5 - 426523.0	7.54863	4.85873	3.88502
Venus	336162.9 - 403952.1	6.43715	4.17518	0.39057
Earth	341081.7 - 399527.9	4.64805	3.08577	3.18475
Mars	351995.4 - 392887.5	4.06628	2.54517	2.64623
Jupiter	374031.0 - 379341.5	0.31678	0.20824	0.21181
Saturn	373178.7 - 375051.0	0.08008	0.08807	0.08908
Uranus	376081.7 - 376232.1	0.03608	0.00728	0.00746
Neptune	370877.1 - 371081.6	0.00487	0.00194	0.00214

The body-Sun subtraction changes both magnitude and direction. In particular, the Sun is not a passive origin: its own spin-coupled K acceleration must be removed. This is why the correct heliocentric observable is $q_i = k_i - k_{\text{Sun}}$ rather than k_i alone.

13. Predicted orbital observables

13.1 Semimajor-axis modulation

The osculating semimajor axis responds one quarter cycle after the tangential force. For complete orbits, the fixed inertial component closes to nearly zero net work. A position or velocity initial-condition refit can absorb much of this periodic term. The non-fixed annual component of the CMB vector produces sidebands and incomplete closure that cannot be represented by one static eccentricity vector.

13.2 Eccentricity and apsidal modulation

The eccentricity vector is the principal retained in-plane state. Its direction sets perihelion, and its negative sets aphelion. A K-force template generates a forced eccentricity component approximately 90 degrees from the projected force. Existing planetary eccentricities add vectorially, so the observed nominal aphelion need not lie at 90 degrees from the K direction. Earth happens to lie close to the K radial direction in the selected epoch; Mercury and Venus do not.

13.3 Inclination and node

The normal component q_N produces inclination and nodal modulation. It does no first-order work on a circular orbit but supplies an essential discriminant because it cannot be absorbed entirely into an in-plane eccentricity correction. Earth has a substantial normal component in the daily calculation; therefore, an observational fit should solve the full three-dimensional variational equation.

13.4 Frequency structure

For each planet, the dominant fixed-ICRS term appears at the orbital frequency. The addition of the body orbital velocity to V_0 modulates the transformed vector annually in the body frame and generates sidebands. Spin-pole precession and periodic pole terms add much slower or smaller components. A multi-body solution can distinguish the K template because each planet carries a different signed spin vector, orbital frequency, and orbital-plane projection.

14. Comparison with known orbit kinematics

The known Keplerian orbit already contains a large radius variation determined by its fitted eccentricity. The K-induced periodic radius term is not compared by simply searching for an additional maximum radius at the predicted date. The correct comparison is a simultaneous vector fit in which the K template competes with initial position, initial velocity, gravitational parameters, asteroid masses, relativistic terms, and the conventional osculating elements.

Within the screening calculation, the 90-degree phase chain is not an empirical coincidence: it is the required integral relation between a sinusoidal tangential acceleration and its accumulated orbital-energy response. The empirical question is whether a common coupling coefficient applied to all body-specific templates improves the fit to precision ranging and astrometry after all conventional parameters are refitted.

15. Variational equation for a definitive ephemeris fit

Let x_i be the barycentric state and $s_i = \partial x_i / \partial \kappa_F$ the sensitivity to the common coupling parameter. The exact linearized solve-for equation is:

$$d^2 s_i / dt^2 = \sum_j [\partial a_{i, \text{std}} / \partial x_j] s_j + [k_i(t) - k_{\text{SSB}}(t)]$$

For a heliocentric state, the forcing reduces to $q_i(t) = k_i - k_{\text{Sun}}$. For an interbody range $\rho_{ij} = |x_i - x_j|$, the direct observation partial is:

$$\partial \rho_{ij} / \partial \kappa_F = \hat{\rho}_{ij} \cdot [s_i(t) - s_j(t)]$$

A complete solution must integrate these sensitivities with the standard dynamical model and jointly refit the ordinary parameters. The one-year daily ledger supplied with this report is a forcing-template and phase-identification product. It is not a replacement for the full DE440-class normal-equation solution.

16. Reproduction procedure

1. Fix the ICRS Cell3 origin at $O=(0,0,0)$ and load the locked primitive basis, G_{phase} matrix, and V_0 CMB vector.
2. Generate daily heliocentric positions from the stated ephemeris model and transform them to ICRF-equatorial coordinates.
3. Reconstruct SSB positions and velocities by mass weighting the Sun and planetary systems.
4. For each body and epoch, construct the signed spin vector from pole direction and prime-meridian rate.
5. Compute $V_{i, \text{CMB}} = V_0 + v_{i, \text{SSB}}$, $g_i = G_{\text{phase}} V_{i, \text{CMB}}$, and $k_i = (1/3) \omega_i \text{ cross } g_i$.
6. Compute the common mode k_{SSB} and the heliocentric differential $q_i = k_i - k_{\text{Sun}}$.
7. Construct the local R,T,N frame and resolve q_R , q_T , q_N .
8. Integrate da/dt and the linearized Cartesian variational response with zero initial perturbation.
9. Fit orbital-frequency sinusoids to q_T and δa for bodies completing full orbits.

10. Integrate complete fixed-force test orbits and evaluate the induced eccentricity and aphelion direction.
11. Compare the entire multi-body vector template with a precision ephemeris through a single shared κ_F solve-for parameter.

17. Numerical checks and interpretation guards

Check	Result
Daily rows	3294 = 366 epochs x 9 bodies
Unique dates	366
Bodies	Earth, Jupiter, Mars, Mercury, Neptune, Saturn, Sun, Uranus, Venus
Cell3 nodes	27
Cell3 vertices	8
Boundary planes	6 in 3 antipodal families
Inner-planet force-to-size phase	Mercury, Venus, Earth within 0.40 deg of 90 deg
Inner-planet induced-aphelion phase	Mercury, Venus, Earth within 0.022 deg of 90 deg
Origin effect on current uniform tensor	No change to force vectors; origin retained for spatial phase indexing
Common-mode rule	Internal heliocentric observable is $q_i = k_i - k_{\text{Sun}}$

Three interpretation guards are essential. First, maximum total K-force magnitude is not generally the point of maximum orbital-energy transfer; q_T is the controlling index. Second, a fixed inertial K vector does not create monotonic semimajor-axis growth around a closed orbit. Third, an origin lock alone does not define a spatial phase-amplitude modulation. It provides the coordinate reference required once such a law is specified.

18. Principal findings

1. The Cell3 origin is now explicitly fixed at $O=(0,0,0)$ in the ICRS-oriented Cartesian frame.
2. The exact heliocentric K observable is the body-Sun differential $q_i = k_i - k_{\text{Sun}}$.
3. Mercury, Venus, and Earth reproduce the predicted force-to-semimajor-axis quarter-cycle relation at 90.1685, 89.9820, and 89.5997 degrees.
4. Complete fixed-force tests place the induced aphelion direction at 89.9952, 90.0212, and 90.0083 degrees from the projected K-force direction.
5. The fixed force performs essentially zero net work over a closed orbit; persistent structure is carried by eccentricity, apsidal, inclination, and sideband modulation.
6. Earth is nearly phase aligned: nominal aphelion is 2.5062 degrees from the projected K-force direction in the selected interval.
7. The Earth alignment is not universal across the planets; Saturn is also close at 7.8821 degrees, while the remaining bodies are substantially separated.
8. Outer planets require multi-year to multi-decade integrations before the same closed-orbit phase test is resolvable.
9. The definitive coupling estimate must come from a common-parameter, multi-body variational fit to integrated precision ephemerides.

19. Conclusion

The volume-filling, ICRS-oriented K-Field calculation produces a precise and reproducible phase architecture in Solar-System dynamics. Each rotating body experiences a body-specific acceleration determined by its signed spin vector and its CMB velocity. The common acceleration of the Solar System cancels from internal dynamics; the body-Sun difference is the orbit-driving term. When this term is projected onto the orbital tangent, its integral produces a semimajor-axis response one quarter cycle later. The computed inner-planet lags confirm that relation. Over a complete orbit, the fixed component returns the semimajor axis while rotating or translating the eccentricity vector, placing the induced aphelion approximately 90 degrees from the projected force. The correct observational target is therefore a three-dimensional, body-indexed variational template rather than a single enlarged-radius event. The Cell3 origin at $O=(0,0,0)$ now fixes the translational gauge for extending the same computation to an explicit spatial phase law.

Appendix A. Complete 27-node Cell3 registry

#	2i	2j	2k	lambda	ICRS X [native]	ICRS Y [native]	ICRS Z [native]
0	-1	-1	-1	-3	-0.010974902500	0.007385983127	0.039661911012
1	-1	-1	0	-2	-0.011290092243	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443559
3	-1	0	-1	-2	-0.009875529516	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653760	-0.020295694327
6	-1	1	-1	-1	-0.008776156532	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292378	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568631	-0.037328945095
9	0	-1	-1	-2	-0.000784183241	0.005677775879	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775879	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568631	0.037328945095
19	1	-1	0	0	0.009091346274	-0.008159292378	0.015866767810
20	1	-1	1	1	0.008776156532	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653760	0.020295694327
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529516	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443559
25	1	1	0	2	0.011290092243	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902500	-0.007385983127	-0.039661911012

Appendix B. Vertices and boundary-plane equations

#	a coeff	b coeff	c coeff	ICRS X	ICRS Y	ICRS Z
0	-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
2	-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
5	0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
7	0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

Family	Plane sign	Unit normal ICRS	d [native]	Equation
ab faces (constant c coefficient)	+	[0.1940922008, 0.9131771688, 0.3583792348]	-0.018828573500	$n \cdot x + (-0.018828573500) = 0$
ab faces (constant c coefficient)	-	[0.1940922008, 0.9131771688, 0.3583792348]	0.018828573500	$n \cdot x + (0.018828573500) = 0$
ac faces (constant b coefficient)	+	[0.0890525941, 0.8665767693, -0.4910339483]	0.014052166903	$n \cdot x + (0.014052166903) = 0$
ac faces (constant b coefficient)	-	[0.0890525941, 0.8665767693, -0.4910339483]	-0.014052166903	$n \cdot x + (-0.014052166903) = 0$
bc faces (constant a coefficient)	+	[-0.9959651903, 0.08360216251, -0.03261929056]	-0.010254361619	$n \cdot x + (-0.010254361619) = 0$
bc faces (constant a coefficient)	-	[-0.9959651903, 0.08360216251, -0.03261929056]	0.010254361619	$n \cdot x + (0.010254361619) = 0$

Appendix C. Complete body summary

Body	Coverage	Force-size lag	Induced-aphelion lag	Nominal aphelion to K	K excursion	CMB excursion	Max radial at bound
Mercury	1498.66094	90.16850	89.99519	165.97002	3.885017	7.548634	1.415169e-04
Venus	583.72239	89.98201	90.02117	139.44063	0.390570	6.437150	5.284392e-04
Earth	359.75536	89.59966	90.00835	2.50620	3.184745	4.648050	0.0226195
Mars	171.43696	106.15493	88.72337	114.57369	2.646234	4.066279	0.0250993
Jupiter	28.77707	103.75155	72.19830	76.35372	0.211813	0.316778	0.0706689
Saturn	12.55167	72.35860	93.62209	7.88206	0.089082	0.080081	0.0238905
Uranus	4.17774	84.89511	73.13204	26.77792	0.007461	0.036081	0.0179095
Neptune	2.21155	88.32135	93.99923	45.65056	0.002138	0.004871	0.0137818

Appendix D. Selected daily vector ledger

Date	Body	V_CMB RA	V_CMB Dec	K RA	K Dec	q	q_R	q_T	q_N
2026-07-23	Earth	164.876	-6.414	276.703	1.117	18.8082	15.8315	-6.5443	7.7643
2026-07-23	Jupiter	169.257	-7.153	272.004	-25.453	52.5123	-43.6625	29.1029	-2.0373
2026-07-23	Mars	166.104	-5.414	269.235	-25.621	22.4569	-16.9232	-14.7568	-0.3918
2026-07-23	Mercury	161.448	-4.290	95.096	23.725	0.5276	-0.3475	0.3946	-0.0432
2026-07-23	Neptune	167.182	-6.600	263.298	-41.395	32.9849	-3.8476	-30.8789	-10.9403
2026-07-23	Saturn	166.679	-6.306	279.680	3.829	49.1969	0.8680	-44.0844	21.8208
2026-07-23	Uranus	167.743	-6.637	318.153	63.759	31.0329	7.0958	-7.3665	29.2989
2026-07-23	Venus	169.209	-9.017	91.345	25.147	0.9489	-0.8320	-0.4560	0.0141
2026-10-23	Earth	164.969	-5.077	277.987	0.961	21.7327	-6.9338	-18.5782	8.8928
2026-10-23	Jupiter	169.405	-7.257	272.137	-25.449	52.3541	-39.5233	34.2746	-2.0270
2026-10-23	Mars	168.754	-6.500	271.054	-26.390	23.1053	-23.0028	2.0582	-0.6977
2026-10-23	Mercury	161.248	-3.531	95.730	23.605	0.5117	-0.2263	0.4569	-0.0432
2026-10-23	Neptune	167.184	-6.599	263.298	-41.394	32.9922	-4.1490	-30.8471	-10.9422
2026-10-23	Saturn	166.711	-6.300	279.720	3.824	49.2982	-1.5147	-44.1600	21.8611
2026-10-23	Uranus	167.760	-6.643	318.157	63.755	31.0394	6.9613	-7.4989	29.3045
2026-10-23	Venus	164.501	-4.580	90.955	25.110	0.9625	0.4451	0.8533	0.0140
2027-01-23	Earth	170.850	-7.470	282.048	0.891	23.1186	-19.9825	6.9488	9.3210
2027-01-23	Jupiter	169.527	-7.355	272.253	-25.446	52.1673	-34.8041	38.8077	-2.0161
2027-01-23	Mars	170.596	-7.663	272.472	-26.969	22.4780	-15.6085	16.1502	-0.8999
2027-01-23	Mercury	161.622	-2.994	96.180	23.419	0.4913	-0.0781	0.4831	-0.0432
2027-01-23	Neptune	167.186	-6.598	263.297	-41.393	32.9996	-4.4501	-30.8123	-10.9441
2027-01-23	Saturn	166.748	-6.296	279.762	3.819	49.3999	-3.9151	-44.1061	21.9014
2027-01-23	Uranus	167.777	-6.649	318.161	63.752	31.0457	6.8240	-7.6290	29.3098
2027-01-23	Venus	172.955	-9.218	91.598	25.122	0.9598	0.0725	-0.9570	0.0137
2027-04-23	Earth	171.155	-8.950	281.231	1.032	20.0887	6.2606	17.2527	8.1675
2027-04-23	Jupiter	169.621	-7.443	272.347	-25.444	51.9606	-29.7195	42.5750	-2.0051
2027-04-23	Mars	171.049	-8.439	273.059	-27.205	21.1378	-1.8803	21.0333	-0.9324
2027-04-23	Mercury	162.069	-2.848	96.282	23.297	0.4795	0.0040	0.4775	-0.0432
2027-04-23	Neptune	167.188	-6.597	263.297	-41.392	33.0069	-4.7443	-30.7755	-10.9459
2027-04-23	Saturn	166.787	-6.294	279.806	3.814	49.4995	-6.2737	-43.9255	21.9406
2027-04-23	Uranus	167.794	-6.655	318.165	63.749	31.0516	6.6871	-7.7538	29.3149
2027-04-23	Venus	163.197	-5.646	90.915	25.139	0.9504	-0.6133	0.7258	0.0144
2027-07-23	Earth	164.889	-6.423	276.710	1.116	18.8055	15.8580	-6.4736	7.7628
2027-07-23	Jupiter	169.690	-7.525	272.423	-25.441	51.7309	-24.2017	45.6770	-1.9939
2027-07-23	Mars	169.778	-8.491	272.447	-26.966	19.5973	11.9053	15.5469	-0.7834
2027-07-23	Mercury	163.087	-2.831	96.192	23.081	0.4607	0.1289	0.4402	-0.0432
2027-07-23	Neptune	167.190	-6.596	263.297	-41.391	33.0142	-5.0416	-30.7354	-10.9478
2027-07-23	Saturn	166.830	-6.294	279.851	3.808	49.5999	-8.6618	-43.6120	21.9799
2027-07-23	Uranus	167.811	-6.661	318.168	63.746	31.0574	6.5460	-7.8776	29.3199
2027-07-23	Venus	170.347	-6.730	91.399	25.099	0.9691	0.9362	-0.2500	0.0136

The complete 3,294-row ledger is contained in the machine-readable companion and the source CSV.

Appendix E. Formula ledger

Object	Formula
Cell3 embedding	$x = O + i \ a + j \ b + k \ c$; $O = [0, 0, 0]_{\text{ICRS}}$
Body CMB velocity	$V_i, \text{CMB} = V_0 + v_i, \text{SSB}$
Transformed vector	$g_i = G_{\text{phase}} V_i, \text{CMB}$
Whole-body coefficient	$k_i = (1/3) \ \omega_i \ \text{cross} \ g_i$
Common mode	$k_{\text{SSB}} = \text{sum}(M_i \ k_i) / \text{sum}(M_i)$
Heliocentric differential	$q_i = k_i - k_{\text{Sun}}$
Orbital projections	$q_R = q \ \text{dot} \ R$; $q_T = q \ \text{dot} \ T$; $q_N = q \ \text{dot} \ N$
Energy rate	$d \ \epsilon / dt = v \ \text{dot} \ (kappa_F \ q)$
Semimajor-axis rate	$da/dt = (2a^2/\mu) \ v \ \text{dot} \ (kappa_F \ q)$
Near-circular phase response	$\delta a = (2 \ kappa_F \ q_{\text{parallel}}/n^2) \ \cos(\theta)$
Mean eccentricity-vector rate	$\langle de/dt \rangle = -[3 \ kappa_F / (2na)] \ \hat{h} \ \text{cross} \ q_{\text{parallel}}$
Variational equation	$d^2 s_i / dt^2 = \text{sum}_j (A_{ij} \ s_j) + [k_i - k_{\text{SSB}}]$

Appendix F. Source references

- [R1] JPL Solar System Dynamics. Approximate Positions of the Planets. Keplerian elements and rates valid for 1800-2050; J2000 ecliptic-to-ICRF conversion; lower-accuracy status and Horizons recommendation. https://ssd.jpl.nasa.gov/planets/approx_pos.html
- [R2] Park, R. S., Folkner, W. M., Williams, J. G., and Boggs, D. H. The JPL Planetary and Lunar Ephemerides DE440 and DE441. *Astronomical Journal* 161:105, 2021. https://ssd.jpl.nasa.gov/doc/de440_de441.html
- [R3] JPL Solar System Dynamics. Astrodynamical Parameters. DE440 planetary-system gravitational parameters. https://ssd.jpl.nasa.gov/astro_par.html
- [R4] NASA Navigation and Ancillary Information Facility. Generic Planetary Constants Kernel pck00011.tpc and associated ICRF pole and rotation definitions. https://naif.jpl.nasa.gov/pub/naif/generic_kernels/pck/
- [R5] Source-locked K-Field continuous-rim machine model: K-Field_flywheel_force_analysis_1784808116.json. Provides G_{phase} , V_0 , Cell3 basis, 27 nodes, vertices, and boundary planes.
- [R6] Daily computation source: K-Field_solar_system_daily_orbit_modulation_2026-07-23_to_2027-07-23.csv and K-Field_solar_system_daily_orbit_modulation_summary.json.

Appendix G. Machine-readable reconstruction metadata

Field	Value
PDF filename	K-Field_solar_system_coupling_analysis_phase_shifted_1784823303.pdf
JSON filename	K-Field_solar_system_coupling_analysis_phase_shifted_1784823303.json
Epoch	1784823303
Generated UTC	2026-07-23T16:15:03Z
Daily source SHA-256	5f48166163941b33d62ede922fa0ca38cf55afb2e9d860b0992755cf70be9628
Summary source SHA-256	2d8ef39cec5cd19a877f09a3929779a76fb247aac673559533eff73128b682d0
Geometry source SHA-256	a965becf29845009e05ca992befb7699a5869415b5296c315bdf81fef10fa4e8
Analysis script SHA-256	dbf84af0efb5de94cf5db1d2913f51f82e860fba433460998d2b30167b7d52
Enrichment script SHA-256	f74cbf0a96a74b4b768996836672b94a37e894fe118607817302173aebab888a